Exp-4.

Text Preprocessing & Analysis

AIM:

To clean and preprocess Amazon reviews, then identify the most frequent words.

PROCEDURE:

1. Load the dataset (amazon-reviews.csv)
2. Clean text: lowercase, remove punctuation/chars, tokenize
3. Remove stopwords tokens.
4. Get cleaned tokens
5. Flatten all tokens, and analyze frequency

CODE:

```
import pandas as pd, spacy
from collections import Counter
df=pd.read_csv("amazon_reviews.csv")
nlp=spacy.load("en_core_web_sm")
def clean(text):
    if pd.isnull(text): return
    doc =nlp(str(text).lower())
    return [t.text for t in doc if not t.is_stop and t.is_alpha]
df["cleaned"]=df["reviewText"].apply(clean)
all_tokens=[tok for tokens in df["cleaned"] for tok in tokens]
freq = Counter(all_tokens)
print("Top 10 frequent words:", freq.most_common(10))
```

RESULT:

Thus the text preprocessing is successfully pipelined and cleaned that also reflects key terms from reviews

OUTPUT:

Top 10 frequent words:

```
[('good',134),
 ('quality',112),
 ('product', 105),
 ('use', 98),
 ('great', 95),
 ('easy', 89),
 ('one',80),
 ('price', 77),
 ('like', 74),
 ('works',70)]
```